

7. Conclusion

Development of the Smart Event Management and Ticket Booking System was successfully completed to systemize the management of events, bookings, payments and users in a single and user-friendly platform. The system is a single Web application that integrates event management, booking management, payment management, authentication and administration functions.

This project incorporates the following technologies: Cloud storage: MongoDB Atlas Frontend: React.js and the use of Vite Backend: Node.js and the use of Express.js. JWT and bcrypt is used to ensure security for authentication, where as role based authorization is used to ensure that the proper option for accessing the functions to the users and the administration based on their roles.

The system was tested using unit testing, integration testing, and system testing to ensure the proper functioning and communication of the various components. It was also successfully configured using Vercel and Railway, to be used as a cloud-based system.

Overall, the project has been completed to a satisfactory level and lays a solid foundation for a robust and scalable event management and ticket booking platform.

7.2 Limitations of the system

The system has limitations:

While it offers fundamental capabilities for managing events and reserving tickets, there are a number of restrictions to consider with the system:

1. Payment process is simulated and not yet connected to a real payment gateway.
2. There is no facility for verification of payment in real time.
3. Ticket verification by QR code is not possible.
4. There is no integration for email and SMS notification.
5. Event analytics & reporting are not advanced.
6. At this time, there is no dedicated app to use.
7. There is no implementation of the AI based event recommendations.
8. No extensive load and stress testing done.

To further develop and integrate the additional services the limitations of the system would have to be overcome and a better application system created.

7.3 future scope of the project

The system can be further enriched by incorporating various features to make the system more user friendly and practical to be implemented in real life application.

Real Payment Gateway Integration

You can integrate a payment gateway like Razorpay or Stripe to enable secure payments on the website and return instant verification of transactions.

QR Code Ticket Verification

Unique QR codes can be created for confirmed bookings – scanned at entry to the event to validate tickets and avoid double or unauthorized access.

Email and SMS Notifications

Automated messages can be set to be sent to users when they register, book an event, receive payment, are reminded of an event, or when it is cancelled or other event-related messages are sent.

Search and Filtering is up to Advanced.

Advanced filters can be given for users such as category, location, date, price, popularity, and availability of tickets.

Seat Selection

A visual seat selection system may be designed to enable passengers to select desired seats prior to confirmation; which seats they select to desire is judged by how the seats are displayed. A visual system of seat selection can be designed to give users the ability to select seats they prefer to book before booking, judged on how the seat is viewed.

Event Analytics

Administrators can be given more detailed statistics on ticket sales, revenue, bookings, popularity of the events and user activity to allow for improved management and decision making.

Mobile Application

Special Android and iOS applications can be created to deliver a main user interface for discovering, booking, payments and issuing electronic tickets.

AI-Based Recommendations

AI can suggest events to users when these are prized, browsing history, and also previous bookings.

Real-Time Notifications

Push notification can be activated to let users know that the events they are interested in are coming up, modified, canceled, booked or other activities.

They can be adapted for implementation in future, to develop a more complete, scalable and commercially viable event management system which is not part of the scope of the existing academic project.